

Eunji KONG

PhD | Neuroscience | Bioimaging

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EDUCATION & RESEARCH

Current Nov 2022	Postdoctoral Research Scientist > Columbia University, USA > Mortimer B. Zuckerman Mind Brain Behavior Institute > Advisor : Drs. Attila Losonczy and Stefano Fusi
Oct 2022 Mar 2022	Postdoctoral Research Scientist > Institute for Basic Science (IBS), South Korea > Center for Cognition and Sociality > Advisor : Dr. Doyun Lee (https://doyunleelab.wordpress.com)
Feb 2022 Sep 2016	Ph.D. > Korea Advanced Institute of Science and Technology (KAIST), South Korea > Graduate School of Medical Science and Engineering > Advisor : Dr. Pilhan Kim (https://ivmvl.kaist.ac.kr) > Dissertation : Development of intravital deep tissue imaging system and investigation of hippocampal mechanism for individual recognition
Aug 2016 Feb 2012	B.S. > Korea Advanced Institute of Science and Technology (KAIST), South Korea > Department of Biological Sciences



HONORS AND AWARDS

2026-2031	K99/R00 Pathway to Independence Award (1K99NS148806-01), NIH/NINDS
Mar 2026	Presenters Travel Grant for Cosyne 2026, Wu Tsai Neurosciences Institute - Stanford University
2024-2025	Overseas Postdoctoral Fellowship, Korea National Research Foundation
Sep 2022	Future of Science Fund Scholarship, Keystone Symposia - Neurocircuitry of Social Behavior
May 2022	Best Presentation Award, Korea Society of Brain and Neural Science Annual Meeting
Feb 2019	Travel Grant Award, Max Planck Florida Institute for Neuroscience - Neuroimaging Course
Nov 2016	Young Investigator Best Research Award, Annual Biophotonics Conference
Oct 2016	Best Paper Award, International Biomedical Engineering Conference
Aug 2016	Encouragement Award, KAIST Undergraduate Research Program Workshop



PUBLICATIONS

Under Review

- [18] Mihaila T, **Kong E**, Negrean A, Geiller T, Peterka D, Losonczy A, "Rapid Photon-Resolved Excitation-Denoised 3-Photon Imaging Improves Detection of Neuronal Activity in Awake and Behaving Mice", **bioRxiv**, 2026
- [17] Fallah K, Chen H, Singha R, **Kong E**, Turi G, Losonczy A, Zabeh E, "NeuralFieldManifold : Reconstruction of LFP manifold with Lag Embedding", **NeurIPS 2026**

Published

- [16] **Kong E^{*,†}**, Zabehe E^{*}, Liao Z^{*}, Mihaila T, Peterka D, Geiller T[†], Losonczy A[†], “Recurrent connectivity shapes spatial coding in hippocampal CA3 subregions”, **Neuron (In Press)**, 2026 (* co-first, † co-corresponding)
- [15] **Kong E**, “How artificial agents can help us understand social recognition”, **The Transmitter** (Postdoc Perspectives), 2026
- [14] Jeon J, Park Y, Kim S, **Kong E**, Kim J, Yang J, Lee J, Kim Y, Kim I, Kim P, “Deciphering perivascular macrophages and microglia in the retinal ganglion cell layers”, **Frontiers in Cell and Developmental Biology**, 12:1368021, 2024.
- [13] **Kong E^{*}**, Lee K^{*}, Do J, Kim P, Lee D, “Dynamic and stable representations of social identity and reward expectancy in the dorsal CA1 hippocampus in male mice”, **Nature Communications**, 14, 2597, 2023. (* co-first)
- [12] Choi J, Choi M, Jeon J, Moon J, Lee J, **Kong E**, Lucia S, Hong S, Lee J, Lee E, Kim P, “*In vivo* longitudinal 920nm two-photon intravital kidney imaging of a dynamic 2,8-DHA crystal formation and tubular deterioration in the adenine-induced chronic kidney disease mouse model”, **Biomedical Optics Express**, 14(4), 1647-1658, 2023.
- [11] Lee E, Lee E, **Kong E**, Jeon H, Park B, Kang N, Yoo J, Lee H, Kim H, Park S, Nam D, Lee J, Park C, Kim P, Lee W, Kim I, “Agonistic Tie2 antibody suppresses normal-to-tumor vascular transition in glioblastoma invading zone”, **Experimental Molecular Medicine**, 55, 470-484, 2023.
- [10] Jeon J, Kim S, **Kong E**, Kim S, Yang J, Lee J, Lee J, Kim Y, Kim P, “Establishment of the reproducible branch retinal artery occlusion mouse model and intravital imaging of the retinal microglia in ischemic reperfusion injury”, **Frontiers in Medicine**, 9:897800, 2022.
- [9] Hong S, Lee J, Moon J, **Kong E**, Jeon J, Kim H, Kim P, “Intravital longitudinal cellular visualization of oral mucosa in a murine model based on rotatory side-view confocal endomicroscopy”, **Biomedical Optics Express**, 13(8), 4160-4174, 2022.
- [8] Hong S, Park Y, Lee J, Moon J, **Kong E**, Jeon J, Park J, Kim H, Kim P, “3D visualization of CD11c+ dendritic cells at dentin-pulp interface in whole intact murine tooth”, **International journal of molecular sciences**, 22(23), 12683, 2021.
- [7] Moon J, Lee E, **Kong E**, Kim P, “*In vivo* two-photon imaging of hepatic steatosis and fibrosis in live small animal model”, **Biomedical Optics Express**, 12(12), 7918-7927, 2021.
- [6] Park I, Hong S, Seok J, Lucia SE, Song E, Kim M, **Kong E**, Seo H, Hwang Y, Ahn S, Kim S, Jang D, Lee J, Lee J, Kim P, Jo Y, “Longitudinal intravital imaging of tumor-infiltrating lymphocyte motility in breast cancer models with different PD-L1 expression levels”, **Journal of Breast Cancer**, 24:e40, 2021.
- [5] Jeon J, Hwang Y, Lee J, **Kong E**, Moon J, Hong S, Kim P, “Intravital imaging of circulating red blood cells in the retinal vasculature of growing mice”, **Translational Vision Science & Technology**, 14(4):31, 2021.
- [4] Moon J, **Kong E**, Lee J, Jung J, Kim E, Park S, Kim P, “Intravital longitudinal imaging of hepatic lipid droplet accumulation in a murine model for nonalcoholic fatty liver disease”, **Biomedical Optics Express**, 11(9), 5132-5146, 2020.
- [3] Lee J, **Kong E**, Hong S, Moon J, Kim P, “*In vivo* longitudinal visualization of the brain neuroinflammatory response at the cellular level in LysM-GFP mice induced by 3-nitropropionic acid”, **Biomedical Optics Express**, 11(8), 4835-4847, 2020.
- [2] Ahn J^{*}, **Kong E^{*}**, Choe K, Song E, Hwang Y, Seo H, Park I, Kim P, “*In vivo* longitudinal depth-wise visualization of tumorigenesis by needle-shaped side-view confocal endomicroscopy”, **Biomedical Optics Express**, 10(6), 2719-2729, 2019. (* co-first)
- [1] Kim Y, Hwang K, Ahn J, Seo Y, Kim J, Lee S, Yoon J, **Kong E**, Jeong Y, Jon S, Kim P, Jeong K, “Lissajous scanning two-photon endomicroscope for *in vivo* tissue imaging”, **Scientific Reports**, 9, 3560, 2019.

PATENTS

- [3] Rep. of Korea Patent 10-2186327, “Apparatus and Methods for In Vivo Wide-area Cellular-level Imaging of Biological Deep Tissue”
Kim P (representative inventor), **Kong E**, Ahn J (Nov. 2020)
- [2] PCT/KR2019/013128, “System for In Vivo Microscopic Imaging of Deep Tissue, and Microscopic Imaging Method”
Kim P (representative inventor), **Kong E**, Ahn J (Oct. 2019)
- [1] KR-10-2016-0070908, “System for Tuberculosis Diagnosis”
Kim Y (representative inventor), **Kong E**, Hyun J, Lee J, Jeon S, Kim H (June. 2016)

SELECTED PRESENTATIONS

- Mar 2026 **Cosyne 2026 | Lisbon, Portugal | Poster**
Kong E^{*}, Zabeh E^{*}, Losonczy A, “How Connectivity Statistics Shape Neural Geometry : Modeling and Causal Validation in Hippocampal CA3”
- Mar 2025 **Cosyne 2025 | Montreal, Canada | Poster**
Kong E^{*}, Zabeh E^{*}, Jacobs J[†], Losonczy A[†], “The recurrency level is a key determinant of representational drift”
- Dec 2024 **Global Young Neuroscientist Symposium 2024 | Daegu, Korea | Invited Talk**
“Recurrent connectivity shapes spatial coding in hippocampal CA3 subregions”
- Oct 2024 **Society for Neuroscience 2024 | Chicago, USA | Poster**
Kong E, Zabeh E, Liao Z, Mihailia T, Peterka D, Losonczy A, Geiller T, “Recurrent connectivity captures distinct spatial memory features along the proximodistal axis of hippocampal CA3”
- Nov 2023 **Society for Neuroscience 2023 | Washington, D.C., USA | Poster**
Kong E^{*}, Geiller T^{*}, Peterka D, Losonczy A, “Coding and dynamics of neural ensembles along the proximo-distal axis of hippocampal CA3 area”
- Nov 2023 **Society for Neuroscience 2023 | Washington, D.C., USA | Poster**
Kong E, Lee K, Do J, Kim P, Lee D, “Dynamic and stable hippocampal representations of social identity and reward expectation support associative social memory in male mice”
- Sep 2022 **Keystone Symposia - Neurocircuitry of Social Behavior | Daejeon, Korea | Talk**
Kong E, Lee K, Do J, Kim P, Lee D, “Dynamic neural representation of social identity in the hippocampus”
- Aug. 2016 **KAIST Undergraduate Research Program Workshop | Daejeon, Korea | Talk**
Kong E, Ahn J, Ahn S, Kim P, “Intravital confocal microscopy based deep brain imaging by CLARITY tissue clearing technique and side-view endomicroscopy”

PROFESSIONAL EXPERIENCE

- Feb 2025 **CCN software workshop, FLATIRON INSTITUTE, USA**
 > Topic : How to use open-source Python packages developed by Flatiron Institute Center for Computational Neuroscience to analyze and model neural data
- May 2024 **CIMER training, COLUMBIA UNIVERSITY, USA**
 > Facilitator : Dr. Kermin J. Martínez-Hernández
 > Topic : Optimizing mentoring relationships
- Feb 2022 **Student Intern, INSTITUTE FOR BASIC SCIENCE, Korea**
Aug 2016 > Center for Cognition and Sociality
 > Advisor : Dr. Doyun Lee (<https://doyunleelab.wordpress.com>)
 > Project : Investigation of neural mechanism underlying individual recognition in the hippocampus
- July 2021 **Research Scientist, LVIS, Korea**
Sep 2021 > Data Science Team (<https://lviscorp.com/>)
- Feb 2019 **Neuroimaging Techniques Winter 2019, MAXPLANCK FLORIDA INSTITUTE FOR NEUROSCIENCE, USA**
Mar 2019 > Advisor : Prof. Yi Zuo (<https://www.zuolab.org/>)
 > Project : *In vivo* structural and functional imaging
- June 2018 **Innovation Workshop, TECHNICAL UNIVERSITY OF DENMARK, Denmark**
July 2018 > KAIST-DTU-NTU-UQ Joint Program
 > Advisor : Prof. Kristian Mølhave
 > Project : Characterization of aerosols for human toxicological assessment

TEACHING AND OUTREACH

Fall 2024 - current	Co-organizer Columbia Hippocampus Club > Co-organizes a monthly seminar series focused on understanding hippocampal evolution and function across through studies of disease, anatomy, activity, and computation.
Fall 2024 - Spring 2025	Teaching Assistance BIOL UN3500 Dept. Biological Sciences, Columbia University > Undergraduate independent biological research > I have provided 12 hours per week of teaching and mentoring for an undergraduate student conducting independent research in the laboratory of Dr. Attila Losonczy.
Summer 2025	Mentoring SURF 2025 Dept. Biological Sciences, Columbia University > Summer undergraduate research program > I have mentored a SURF undergraduate student in the Losonczylab, investigating the effects of aging on the dynamics of place field formation by performing in vivo two-photon calcium imaging in the hippocampus of freely moving animals.
Summer 2024	Mentoring BRAINYAC program Zuckerman Institute, Columbia University > Summer research program for high school students > I have mentored a BRAINYAC high school student in the Losonczylab, teaching immunohistochemistry, tissue imaging, and image processing with Python. The mentee was awarded the Merit Fellowship for outstanding performance.
2023 - current	Volunteering Saturday Science Zuckerman Institute, Columbia University > Hands-on brain science event series for students, families and local community groups > I designed and coordinated the "Where am I" activity section, integrating a virtual-reality spatial navigation system in the Losonczylab to educate the local community about how place cells in the brain work, receiving high evaluations for engagement and effectiveness.
2023 - 2024	Professional service Art in the Ed Lab Zuckerman Institute, Columbia University > A collaborative science-art program that aimed at creating artworks that inspire community members to explore science through an artistic lens. > I participated as an artist selection committee member and discussed my research projects with selected artists to inspire the creation of neuroscience-related artwork.